

Simulate and Analyze a Digital Communication System using Matlab

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Abstract—This project presents the design and simulation of a digital communication system using Matlab. The simulated system employs Quadrature Phase Shift Keying (QPSK) modulation and demodulation, along with Low-Density Parity-Check (LDPC) coding and decoding. We analyze the system's performance by evaluating the Bit Error Rate (BER) over Additive White Gaussian Noise (AWGN) and Rayleigh fading channels.

Index Terms—QPSK, LDPC, AWGN, Rayleigh Fading, BER, Matlab.

I. INTRODUCTION

Digital communication systems play a crucial role in modern wireless and wired networks, enabling reliable transmission of information over noisy and fading channels. However, channel impairments such as thermal noise, multipath propagation, and signal attenuation significantly degrade transmission quality and increase the probability of bit errors.

To improve communication reliability, modern systems employ advanced modulation and channel coding techniques. Quadrature Phase Shift Keying (QPSK) is one of the most widely used digital modulation schemes due to its high spectral efficiency, where each transmitted symbol carries two information bits. In addition, Low-Density Parity-Check (LDPC) codes provide powerful forward error correction capabilities and are adopted in many modern communication standards, including Wi-Fi, DVB-S2, 5G NR, and satellite communications.

The objective of this project is to design and simulate a complete digital communication system in Matlab. The system consists of random bit generation, LDPC encoding, QPSK modulation, transmission through Additive White Gaussian Noise (AWGN) and Rayleigh fading channels, QPSK demodulation, and LDPC decoding. System performance is evaluated using Bit Error Rate (BER) as a function of the signal-to-noise ratio (SNR).

The simulation aims to investigate the impact of channel impairments on communication quality and demonstrate the effectiveness of LDPC coding in mitigating transmission errors.

II. METHOD

Fig. 1 illustrates the overall architecture of the proposed communication system. The transmitted information bits are first encoded using LDPC codes and then modulated using QPSK. The modulated symbols are transmitted through a Rayleigh fading channel corrupted by AWGN. At the receiver,

Zero-Forcing (ZF) equalization is applied before QPSK demodulation and LDPC decoding. Finally, the recovered bits are compared with the original transmitted bits to evaluate the Bit Error Rate (BER). The system simulation consists of several key components:

A. Bit Source and LDPC Coding

A random binary sequence of length 10^6 bits is generated to represent the information source. Before transmission, the data are encoded using a Low-Density Parity-Check (LDPC) code.

LDPC codes were first proposed by Robert Gallager in 1962 and are considered one of the most powerful error-correcting codes currently available. They are defined by a sparse parity-check matrix $\mathbf{H}$ containing mostly zeros and only a small number of ones. The sparsity of $\mathbf{H}$ significantly reduces decoding complexity while maintaining excellent error-correction performance.

For an LDPC code, a valid codeword $\mathbf{c}$ must satisfy

$$\mathbf{H}\mathbf{c}^T = 0.$$

At the receiver, decoding is performed iteratively using soft information represented by Log-Likelihood Ratios (LLRs). Compared with hard-decision decoding, soft-decision decoding exploits the reliability information of received symbols and therefore provides substantially better performance.

B. LLR-Based LDPC Decoding

The decoder calculates the Log-Likelihood Ratio (LLR) for every received bit. The LLR is defined as

$$LLR(b) = \ln \left(\frac{P(b=0|y)}{P(b=1|y)} \right),$$

where y denotes the received signal sample.

A positive LLR indicates that the received bit is more likely to be 0, whereas a negative LLR indicates a higher probability that the bit is 1.

In our project, we use a Max-Log-MAP algorithm to approximate the value of llr.

$$LLR(b) \approx \frac{1}{2\sigma^2} \left[\min_{s \in S_i^0} \|y - hs\|^2 - \min_{s \in S_i^1} \|y - hs\|^2 \right] \quad (1)$$

The LDPC decoder employs the Min-Sum Algorithm, which is a simplified version of the Sum-Product Algorithm (SPA).

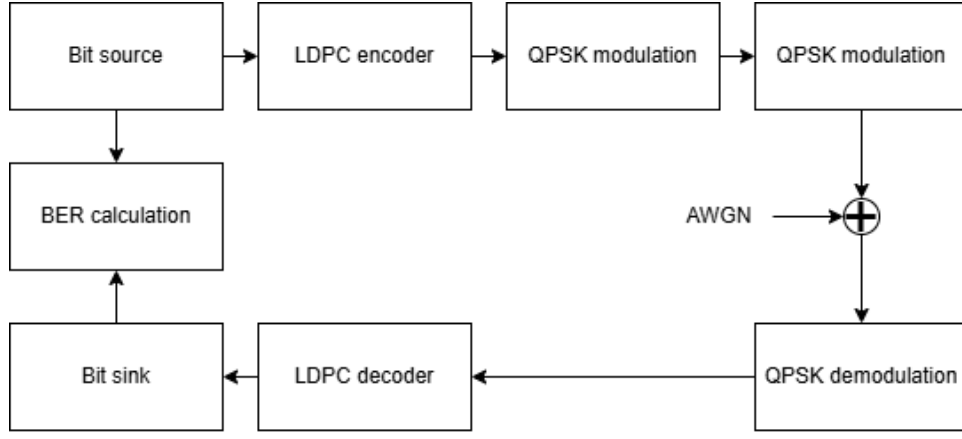


Fig. 1. The block diagram of our communication system

1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	0
0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
1	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	1	0	0	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	1	0	0	0
0	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	1

Fig. 2. Example of a low-density parity-check (LDPC) code matrix for $n = 20$, $j = 3$, and $k = 4$.

Instead of using computationally expensive hyperbolic tangent operations, the Min-Sum Algorithm approximates message updates using sign operations and minimum-value calculations.

For a check node update, the outgoing message is computed as

$$m_{c \rightarrow v} = \left(\prod_{v' \neq v} \text{sign}(m_{v' \rightarrow c}) \right) \cdot \min_{v' \neq v} |m_{v' \rightarrow c}|.$$

This approximation significantly reduces computational complexity while maintaining near-optimal decoding performance.

C. QPSK Modulation

Quadrature Phase Shift Keying (QPSK) is a phase modulation technique in which each symbol represents two information bits. Therefore, the symbol rate is only half of the bit rate, resulting in improved bandwidth efficiency.

The input bit stream is divided into pairs and mapped onto four constellation points located on the complex plane. Gray coding is adopted to minimize the number of bit errors caused by symbol errors.

The mapping used in this project is

$$00 \rightarrow 45^\circ,$$

$$01 \rightarrow 135^\circ,$$

$$11 \rightarrow 225^\circ,$$

$$10 \rightarrow 315^\circ.$$

The transmitted QPSK symbol can be represented as

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t),$$

where $I(t)$ and $Q(t)$ are the in-phase and quadrature components, respectively, and f_c is the carrier frequency.

Using Euler's representation, each symbol can also be expressed as

$$s_k = e^{j\theta_k}.$$

The constellation diagram consists of four equally spaced points separated by 90° , providing robust performance in noisy environments.

D. Rayleigh Fading Channel

In practical wireless systems, transmitted signals often reach the receiver through multiple propagation paths caused by reflection, scattering, and diffraction. Under non-line-of-sight (NLOS) conditions, the channel gain is commonly modeled as a Rayleigh random variable.

The received signal is expressed as

$$y = hx + n,$$

where

- x is the transmitted symbol,
- h is the complex Rayleigh fading coefficient,
- n is additive Gaussian noise.

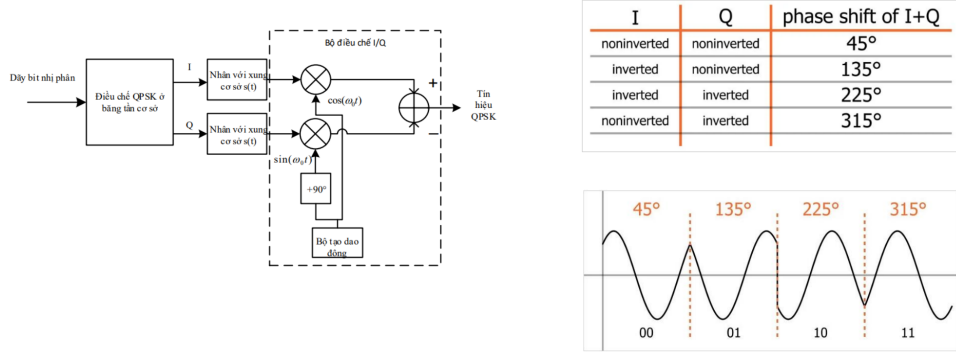


Fig. 3. QPSK modulation using I/Q components, illustrating the modulator structure, phase mapping, and corresponding constellation phases.

Rayleigh fading introduces several impairments:

- Amplitude fluctuations (fading),
- Random phase rotations,
- Doppler frequency shifts caused by mobility.

Deep fades can significantly reduce signal power and dramatically increase the bit error rate.

E. AWGN Channel

The Additive White Gaussian Noise (AWGN) channel is the most fundamental communication channel model and serves as a theoretical benchmark for performance evaluation.

The received signal is given by

$$y = x + n,$$

where n is a Gaussian random variable with zero mean and variance σ^2 :

$$n \sim \mathcal{N}(0, \sigma^2).$$

The term “Additive” indicates that noise is added directly to the signal, “White” means that the noise power is uniformly distributed across all frequencies, and “Gaussian” refers to the probability distribution of the noise amplitude.

Because no fading is present, AWGN represents the best-case scenario for wireless communication systems.

F. QPSK Demodulation and Equalization

At the receiver, the signal first undergoes channel equalization. Since the channel coefficient h is assumed to be known, Zero-Forcing (ZF) equalization is employed:

$$\hat{x} = \frac{y}{h}.$$

This operation compensates for both amplitude attenuation and phase rotation introduced by the Rayleigh fading channel.

The equalized symbols are then converted into soft LLR values and passed to the LDPC decoder. Finally, the recovered bits are compared with the transmitted bits to compute the Bit Error Rate (BER):

$$BER = \frac{N_{error}}{N_{total}}.$$

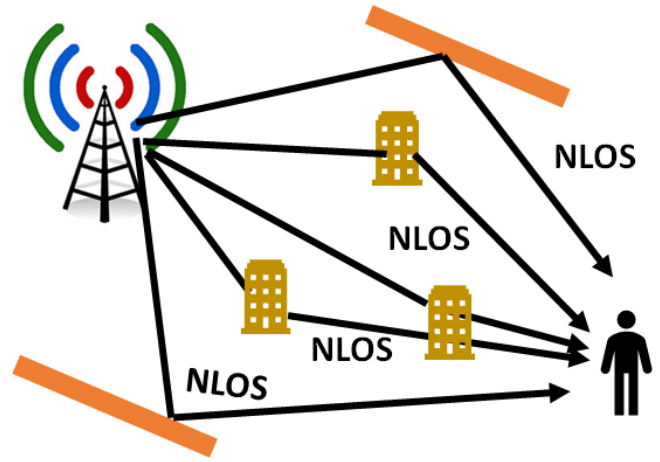


Fig. 4. Illustration of Rayleigh fading channels

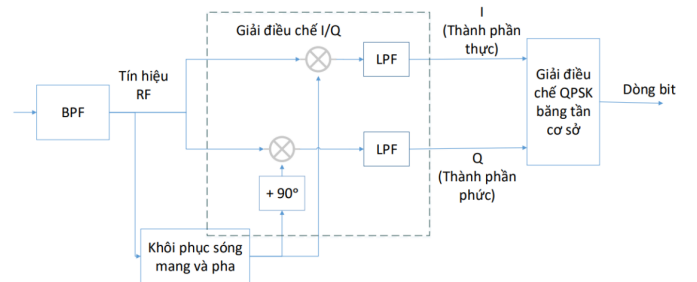


Fig. 5. Illustration of a real-life QPSK demodulation model

III. EXPERIMENTAL RESULTS

The performance of the proposed communication system was evaluated using BER as the primary performance metric. Simulations were performed over a range of E_b/N_0 values for three different scenarios to demonstrate the impact of the channel and the effectiveness of channel coding.

1) Uncoded QPSK over AWGN channel.

As shown in Fig. 6, the simulated BER of the uncoded QPSK signal over an AWGN channel aligns perfectly with the theoretical curve. The BER drops rapidly in

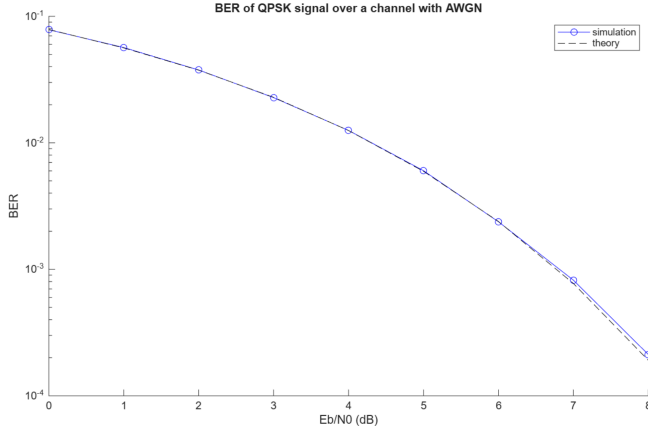


Fig. 6. BER of QPSK signal over a channel with AWGN

a characteristic “waterfall” shape as the signal-to-noise ratio increases. At an E_b/N_0 of approximately 8 dB, the BER decreases to 10^{-4} . This confirms the correct implementation of the base modulation and demodulation under ideal thermal noise conditions.

2) Uncoded QPSK over Rayleigh fading channel.

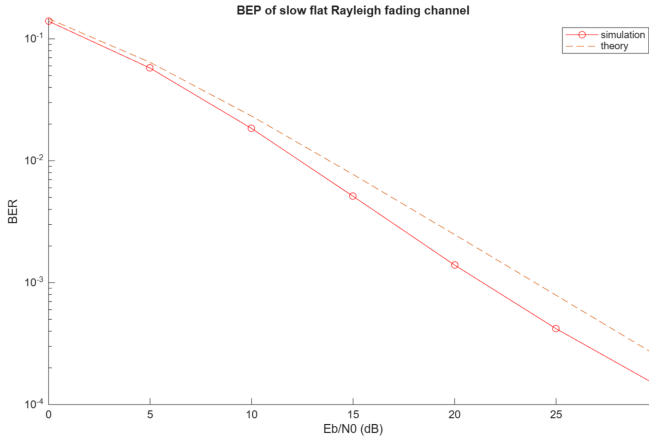


Fig. 7. BER of slow flat Rayleigh fading channel

Fig. 7 illustrates the system’s performance over a slow flat Rayleigh fading channel without any error correction coding. The simulated results tightly follow the theoretical expectation. Unlike the exponential drop observed in the AWGN scenario, the BER curve here decreases linearly on a logarithmic scale, which is a known fundamental characteristic of Rayleigh fading ($BER \propto 1/SNR$). To achieve a BER of 10^{-4} , an E_b/N_0 of nearly 30 dB is required. This highlights the severe performance degradation caused by multipath fading.

3) LDPC-coded QPSK over Rayleigh fading channel.

Fig. 8 demonstrates the significant impact of applying LDPC coding ($N = 1296$, Rate=0.5) to the system over the Rayleigh fading channel. While the coded system performs similarly to the uncoded system at very low

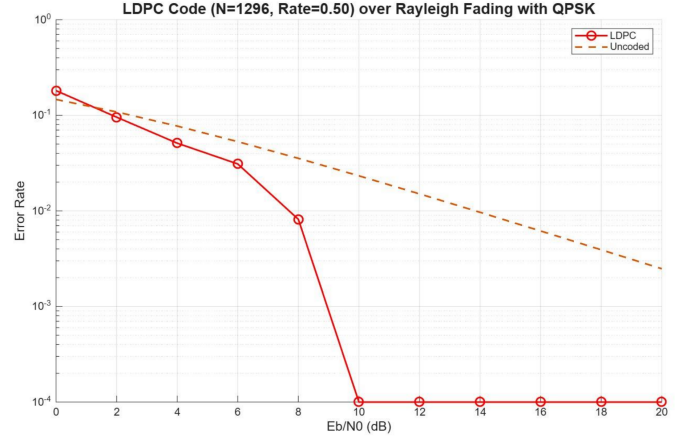


Fig. 8. LDPC Code ($N=1296$, Rate=0.5) over Rayleigh Fading with QPSK

E_b/N_0 values (0 to 2 dB) due to the coding overhead, it exhibits a steep waterfall region starting around 6 dB. The BER plummets dramatically, reaching 10^{-4} at exactly 10 dB E_b/N_0 .

Comparing Fig. 7 and Fig. 8, it is evident that to achieve a BER of 10^{-4} , the uncoded system requires 30 dB, whereas the LDPC-coded system requires only 10 dB. The iterative LDPC decoder successfully corrects most transmission errors caused by fading and noise, providing an impressive coding gain of approximately 20 dB. This demonstrates the strong error-correction capability of LDPC codes and validates their critical role in modern wireless communication systems.

IV. CONCLUSION

This project successfully implemented and simulated a complete digital communication system using Matlab. The system integrated LDPC channel coding, QPSK modulation, AWGN and Rayleigh fading channel models, QPSK demodulation, and LDPC decoding.

Simulation results demonstrate that channel conditions strongly influence communication reliability. While the AWGN channel provides the best baseline performance, Rayleigh fading significantly degrades transmission quality due to multipath propagation effects. The application of LDPC coding greatly improves robustness against channel impairments and achieves near-error-free communication at relatively low SNR values.

The results confirm that the combination of QPSK modulation and LDPC coding provides an effective solution for reliable digital communications in practical wireless environments.

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